

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1707

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I KISHORE KUMAR V(192521371) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

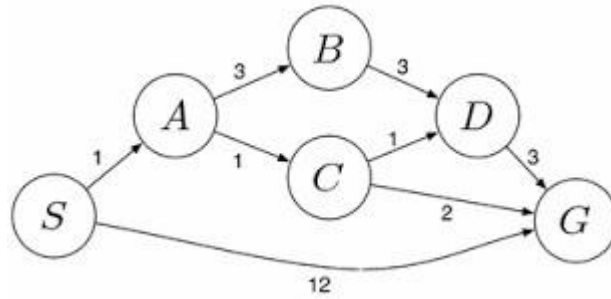
KISHORE
Signature of the student:

Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

1. Water Jug Problem – 4 Gallon and 3 Gallon Jugs

Problem

We have:

- One 4-gallon jug
- One 3-gallon jug
- No measuring marks on either jug
- A water pump
- We can fill a jug completely, empty a jug, or pour water from one jug into another.

The objective is to obtain exactly 2 gallons in the 4-gallon jug.

Solution

Let the state be represented as:

(x, y)

where:

- x = amount of water in the 4-gallon jug
- y = amount of water in the 3-gallon jug

Initial state:

$(0, 0)$

Goal state:

$(2, 0)$

Steps

Step	Operation	4-Gallon Jug	3-Gallon Jug
1	Start with both empty	0	0
2	Fill the 3-gallon jug	0	3
3	Pour 3-gallon jug into 4-gallon jug	3	0
4	Fill the 3-gallon jug again	3	3
5	Pour 3-gallon jug into 4-gallon jug until 4-gallon jug is full	4	2
6	Empty the 4-gallon jug	0	2
7	Pour remaining 2 gallons into 4-gallon jug	2	0

Final Answer

Therefore:

4-gallon jug = 2 gallons

3-gallon jug = 0 gallons

Hence, we successfully obtain exactly 2 gallons in the 4-gallon jug.

2. Mars Rover – Intelligent Agent

The question asks us to analyze a Mars Rover as an intelligent agent that can analyze and explore samples on Mars. The assessment specifically asks about its percepts, environment, actions, performance and suitable architecture.

i. What are the percepts for this agent?

Percepts are the information received by the Mars Rover through its sensors.

The Mars Rover can perceive:

1. **Images of the surrounding environment** using cameras.
2. **Terrain conditions** such as rocks, sand, slopes and obstacles.
3. **Distance from obstacles** using sensors.
4. **Temperature and atmospheric conditions.**
5. **Chemical composition of soil and rocks.**
6. **Position and location** using navigation and positioning systems.
7. **Condition of collected samples** using scientific instruments.
8. **Battery and equipment status.**

Example

If the Rover's camera detects a large rock in front of it, the rock becomes part of its perceptual information.

ii. Characterize the operating environment

The Mars Rover operates in a **complex, partially observable and dynamic environment.**

Characteristics

1. Partially observable

The Rover cannot observe everything on Mars at once. Its sensors only provide information about its nearby surroundings.

2. Dynamic

The Rover's operating conditions can change due to terrain, dust, temperature and other environmental factors.

3. Sequential

A current action can affect future decisions. For example, choosing one path can determine which locations the Rover can visit later.

4. Uncertain

The Rover may not know exactly what is present underneath the soil or beyond an obstacle.

5. Continuous

Movement, distance, temperature and other environmental properties can have continuous values.

6. Single-agent environment

The Rover can primarily be considered a single intelligent agent performing its mission, although communication with Earth and other systems can influence its operation.

Overall characterization

The Mars environment is:

Partially observable + uncertain + dynamic + sequential + continuous

iii. What actions can the agent take?

The Rover can perform several actions:

1. **Move forward**
2. **Move backward**
3. **Turn left**
4. **Turn right**
5. **Stop**
6. **Navigate around obstacles**
7. **Collect soil or rock samples**
8. **Analyze samples**
9. **Capture images**
10. **Transmit information to Earth**
11. **Select another exploration location**
12. **Recharge or conserve energy when required**

Example

If the Rover detects a rock that looks scientifically interesting, it can:

**Detect → Move toward rock → Collect sample → Analyze sample →
Store results → Send data**

iv. How can we evaluate the performance of the agent?

The Rover's performance can be measured using a **performance measure**.

Important performance measures include:

- Amount of useful scientific data collected
- Number of samples successfully collected
- Accuracy of sample analysis

- Distance successfully explored
- Number of scientific discoveries
- Energy consumption
- Time taken to complete missions
- Ability to avoid obstacles
- Successful communication with Earth
- Safety of the Rover

Example

A Rover that collects many useful samples while using less energy and avoiding damage would have **better performance**.

v. Suitable agent architecture and why?

The most suitable architecture is a **Model-Based, Goal-Based and Utility-Based intelligent agent**, with autonomy and learning capabilities.

Why?

A Mars Rover cannot simply react to the current sensor input because it must:

- Remember previous observations.
- Build an internal model of its surroundings.
- Decide which locations to explore.
- Select actions that help achieve mission goals.
- Consider risks such as obstacles and limited energy.
- Choose the most useful scientific samples.

Suitable architecture

Sensors → Perception → Internal Model → Goal/Utility Evaluation → Decision Making → Actuators

For example:

Detect rock → Remember location → Evaluate scientific value → Decide to collect → Move → Collect sample

Conclusion

A **model-based goal/utility-oriented agent** is most appropriate because the Mars Rover must make intelligent decisions using incomplete information while achieving long-term exploration goals.

3. 8-Queens Problem – Search Space and Problem Formulation

The assessment asks us to formulate the **8-Queens problem** as a search problem, where 8 queens must be placed on a chessboard so that no queen attacks another queen.

Problem Definition

We have an **8×8 chessboard**.

We need to place **8 queens** such that:

- No two queens are in the same row.
- No two queens are in the same column.
- No two queens are on the same diagonal.

A queen can attack another queen horizontally, vertically or diagonally.

Problem Formulation

1. Initial State

Initially, **no queens are placed.**

Therefore:

Initial state = empty chessboard

2. State

A state represents a partial arrangement of queens.

For example:

[1, 3, 5]

can represent:

- Queen 1 $\rightarrow$ column 1
- Queen 2 $\rightarrow$ column 3
- Queen 3 $\rightarrow$ column 5

The exact representation can vary, but each state represents a valid partial configuration.

3. Actions

The action is:

Place a queen in a safe square.

A queen can only be placed if it does not attack any previously placed queen.

4. Transition Model

When we place a queen in a safe position, the current state changes to a new state.

Example:

Empty $\rightarrow$ Place Q1 $\rightarrow$ Place Q2 $\rightarrow$ Place Q3 $\rightarrow$... $\rightarrow$ Place Q8

Each step expands the search tree.

5. Goal Test

The goal is reached when:

8 queens are placed and no queen attacks another queen.

Therefore:

Goal = 8 non-attacking queens

6. Path Cost

For the basic 8-Queens problem, every queen placement can have a cost of **1**.

Therefore, the total cost is:

8 placements $\times$ 1 = 8

The main objective is finding a valid solution rather than minimizing a special cost.

Constraints

For every pair of queens:

Same row

Not allowed.

Same column

Not allowed.

Same diagonal

Not allowed.

For two queens at positions:

$(r1, c1)$ and $(r2, c2)$

they are on the same diagonal if:

$$|r1 - r2| = |c1 - c2|$$

Therefore, a valid solution must satisfy all three constraints.

Example of a Valid Solution

One possible arrangement is:

[1, 5, 8, 6, 3, 7, 2, 4]

This means:

Row Column

1	1
2	5
3	8
4	6
5	3
6	7
7	2
8	4

No two queens attack each other.

Search Strategies

The problem can be solved using:

- Breadth-First Search
- Depth-First Search
- Backtracking
- Constraint Satisfaction techniques

Backtracking/DFS is particularly suitable because we can place queens one at a time and backtrack whenever a conflict occurs.

4. OLA Cab Booking – Problem-Solving Agent

The problem describes a customer using an OLA Cab booking application to travel from one location to another. The customer can select different cab types such as Mini, Micro, Sedan, Shared, Prime, etc.

Type of Problem-Solving Agent

A suitable agent is a **Goal-Based Problem-Solving Agent**.

The main goal is:

Transport the customer from the source location to the destination safely and efficiently.

The agent considers available cab types, cost, distance, estimated time and customer preference.

Problem Formulation

Initial State

Customer is at the **source location**.

Example:

Source = Chennai Central

Goal State

Customer reaches the **destination**.

Example:

Destination = Airport

Actions

The application can:

- Select cab type
- Check available cabs
- Calculate fare
- Estimate travel time
- Book the cab
- Track the cab
- Navigate to destination
- Complete the trip

Path Cost

The cost can include:

- Fare
- Travel time
- Distance
- Customer preference

Pseudocode

START

```
Input source location
Input destination location
Find available cabs
Display cab types:
    Mini
    Micro
    Sedan
    Shared
    Prime
For each available cab:
    Calculate estimated fare
    Calculate estimated travel time
    Display details
Customer selects preferred cab
Confirm booking
```

```
Assign nearest available cab
Track cab location
Navigate from source to destination
When destination is reached:
    Complete trip
    Calculate final fare
    Display trip details

STOP
```

Explanation

The agent receives the customer's source and destination. It searches for available cab options and provides information such as price and estimated time. The customer selects a suitable cab, and the system books it. The cab then navigates from the source to the destination.

Thus, the **goal-based agent** is suitable because it takes actions to achieve the specific goal of reaching the destination.

5. Uniform Cost Search – Logistics Delivery Network

The fifth question describes a logistics company that needs to transport packages between warehouses at **minimum cost**. Each route has an associated travel cost, and the objective is to find the least-cost path from starting warehouse **S** to destination **G** using **Uniform Cost Search (UCS)**.

Important note

The uploaded PDF's parsed text does **not include the actual weighted graph shown below the question**. Therefore, I should not invent the warehouse

connections or edge costs. The method below shows exactly how to solve the given graph once its weights are visible.

What is Uniform Cost Search?

Uniform Cost Search (UCS) expands the node having the **lowest total path cost from the starting node**.

It uses a **priority queue**.

The priority is:

$$\mathbf{f(n) = g(n)}$$

where:

- $g(n)$ = total cost from starting node S to node n.

Unlike Breadth-First Search, UCS considers the **cost of each path**, not simply the number of edges.

UCS Algorithm

Step 1

Start from node S.

Set:

$$\mathbf{Cost(S) = 0}$$

Step 2

Put S into the priority queue.

Step 3

Remove the node having the lowest cumulative cost.

Step 4

Expand that node and calculate the costs of its neighboring nodes.

Step 5

Add the new paths to the priority queue.

Step 6

Always select the node with the **smallest cumulative cost**.

Step 7

Continue until the goal node **G** is removed from the priority queue.

Step 8

The path obtained is the **least-cost path**.

Example of UCS Calculation

Suppose the graph contains:

$S \xrightarrow{-2} A$

$S \xrightarrow{-5} B$

$A \xrightarrow{-3} C$

$B \xrightarrow{-1} C$

$C \xrightarrow{-4} G$

Then:

Start

$S = 0$

Expand S:

- $S \rightarrow A = 2$
- $S \rightarrow B = 5$

Priority queue:

A(2), B(5)

Expand A:

- $A \rightarrow C = 2 + 3 = \mathbf{5}$

Queue:

B(5), C(5)

Expand B:

- $B \rightarrow C = 5 + 1 = 6$

Existing cost to C is 5, so we keep:

$$C = 5$$

Expand C:

- $C \rightarrow G = 5 + 4 = 9$

Therefore:

$$S \rightarrow A \rightarrow C \rightarrow G$$

Total cost:

$$2 + 3 + 4 = 9$$

So the least-cost path is:

$$S \rightarrow A \rightarrow C \rightarrow G$$

$$\text{Minimum Cost} = 9$$

